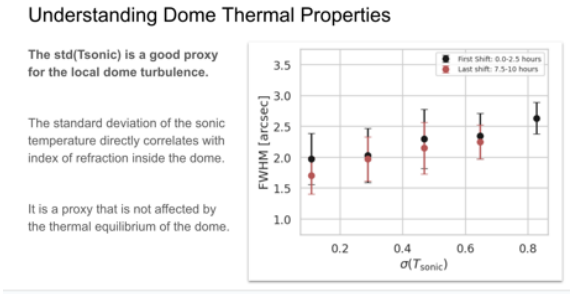
2024 acoustics - Bidirectional Acoustic Traveltime Pulser for Index Gradients (BATPING)!

The sonic anemometer has proven very useful in the determination of temperature fluctuations in the dome air, and that in turn does correlate with FWHM on Aux Tel images.

There are multiple advantages to this basic approach, including-
quantitative rather than qualitative determinations of index of refraction structure.
effect of a temperature change on fractional change in speed of sound is about 500X larger than fractional change on optical index of refraction.
We could do acoustic tomography of 3D volume, as a long term goal.
no light to interfere with observations
no dependence on dome illumination, runs day and night.
Might even repel rodents...

Avoiding multipath means we use short pulses and catch the first echo.
Need to do round trip travel time to disentangle wind from temperature. For now we'll use a corner reflector to get good return intensity.

Speed of sound at 20C changes by 0.606 m/s per degree. That's a fractional change of 0.176% per degree.
Standard deviation of acoustic temperature at 10 Hz in dome is around 0.1 to 0.5 deg C. Plot from Johnny:



That's over a volume of perhaps 20cm in diameter, measured over a 1 sec interval with wind speeds of under 1 m/s. So we can imagine that corresponds to (generously) rms over a 1m path length.
Dependence on distance depends on power spectrum, but if we just take N uncorrelated blobs of air in 1m chunks, we'd expect the standard deviation to scale as sqrt(N).
So over a 10m path the standard deviation might be about 3X larger.

A good initial target is to measure standard deviations of acoustic temperature down to 0.1 C rms, for a round trip path length of 10m.
This is consistent with the numbers in the dome seeing handbook below.

That requires measuring round trip travel time with a fractional uncertainty of better than ~2E-4. How long does it take to go 10 m? dt= 10m/330 m/s ~ 30 msec.
How well do we need to know this? To 30 msec * 2e-4 = 6 microseconds.
For an audio sampling rate of 192 kHz we get sampling interval of ~ 5 microsec, so even faster sampling would be better.
Dodotronics ultramic samples at 384 kHz, so we get Nyquist at 192 kHz.
Imagine Fried length is 20 cm. If wind speed is 1 m/s then how far do things move in 30 msec round trip travel time? dx = 30mm, or about an inch. We are assuming wind vector field is stable over 30 msec.
To avoid multipath contamination what pulse repetition rate can we tolerate? Tens of Hz.

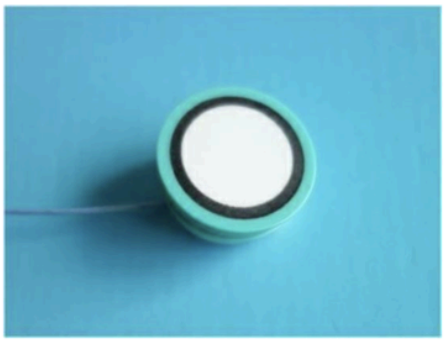
Steminc sells a variety of large format high freq ultrasound piezo parts. Need 100V drive though.

Need to investigate emitted power if these are driven below their resonance.

We can also look into high end audio tweeters, such as Townsend https://www.townshendaudio.com/supertweeters/ goes up to 90 kHz.

Order placed with Steminc Apri 30 2024:

Description	Price	Quantity	Total
SMATR400HPKLR - Air Transducer 400 KHz Long Range - /1 Pc Set	$39.00	2	$78.00
SMATR300H19XDA - Ultrasonic Air Transducer 300 KHz - /1 Pc Set	$41.00	2	$82.00
SMATR175D28PK - Piezo Air Transducer 175 KHz 28mm - /1 Pc Set	$43.00	2	$86.00
		Sub-Total:	**$246.00**
		Shipping:	**$11.99**
		Order Total:	**$257.99**



Piezo Air Transducer 175 KHz
SMATR175D28PK - 1 Pc/Set
Manufacturer part number: SMATR17
Availability:
51 in stock
$43.00 /1 Pc Set
Shipping: $11.99
Shipping Each Additional unit: $0.00
1 ADD TO CART
Condition: New
Email a friend

Part Number: SMATR175D28PK

Our Piezo Air Transducer is Water proof and it is Transmitter/Receiver on the same unit.
2m Ultrasonic Proximity Sensors.

Housing Material: Plastic & Copper
Dimensions: 28mm X 12.3mm
Nominal frequency fr: 175kHz ± 2%
Wires/Connections: 32mm Wires
Sensitivity: -68 dB Min. (0dB = 100Vpp 10 Bursts/20cm)
Capacitance: 600pF±20%
Max. Input Voltage: 150Vp-p (Impulse wave)
Ringing: 0.6 ms Max.
Tipical Directive (-6dB): 10°
Test Condition: 22±3 °C 40~70% R.H.
* Actual properties will depend on driving circuit. Screenshot

Some scalings:
Frequencies above ~100 kHz are highly attenuated.

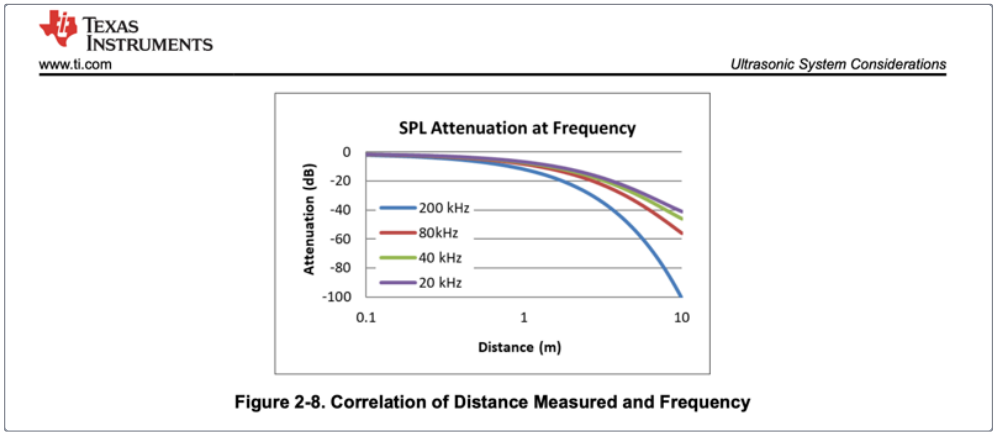
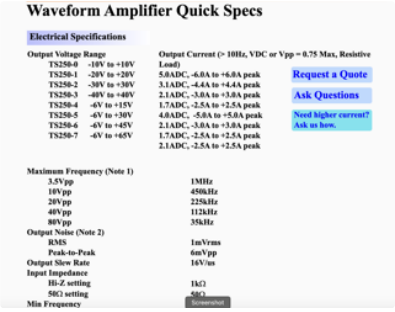


Figure 2-8. Correlation of Distance Measured and Frequency

Acoustic wavelength is 3.3mm X (100 kHz/f).
Emission angle FWHM is lambda/D = 7.4 degrees X (1 inch/D_emitter) X (100 kHz/f) .
Beam size at 10 m is 1.3 meters for 1 inch dia emitter and 100 kHz freq. At 2m distance the beam footprint is ~ 30 cm or about a foot.
Note that 40 kHz is where ultrasonic ranging systems tend to operate.

If capacitance is 600pF = 0.0006 uF, at 175 kHz the impedance is Z=1/2*pi*f*C = 1.5 KOhms.
Current needed at 100V is I=V/Z = 100/1.5E3 = 66 mAmps. If we put a 50 Ohm resistor in series we get tau = 50 * 600E-12 = 30 nsec.

I have a pulse amplifier on hand. Accel TS 250-0. https://www.accelinstruments.com/Products/TS250/Waveform-Amp.html
It delivers 6A rms at 10V peak-peak. So optimal impedance is in fact about 1.5 Ohms. We could drive a LOT of these in parallel.
We can set it to unity voltage gain, 0 dB.

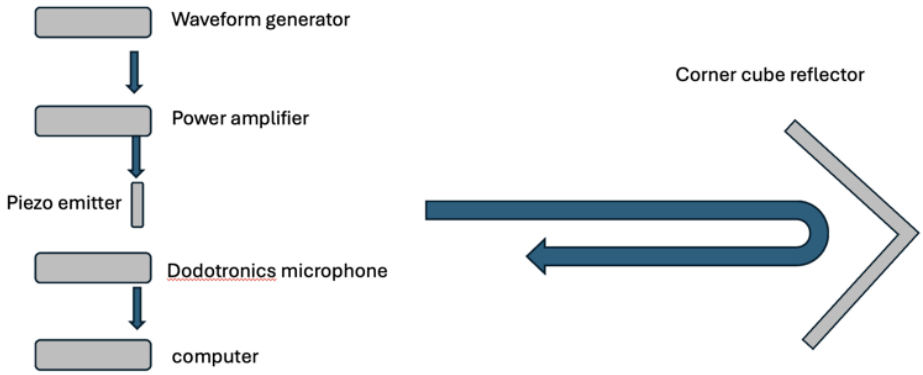


If we want round trip timing resolution of 5 microseconds on a pulse train with carrier freq of 100 kHz, i.e. period of 10 microseconds, we only need to know return phase to within pi radians. Not so hard!
We actually want a higher drive voltage, ideally up around 100V. Could use a transformer to do that. Needs about a 1:10 turns ratio. Ordered a ferrite core to make our own pulse transformer.

parts list

item	vendor	notes
transducers, various resonant freq's	Steminc	ordered, inbound
waveform generator	varies	on hand
current-amplifying buffer amp	Accel	TS-250-0 on hand, drives 6A at 10 V at 450 kHz.
12-inch retroreflector	Amazon	ordered kind used for boats.
microphone	dodotronics	Ultramic 384K EVO

Version 0.5:



Start time comes from microphone picking up the emitted pulse. End time comes from returned echo. Do correlation analysis to find peaks.
Waveform generator could send either Gaussian wave packet or else single cycle, followed by phase-flipped cycle to suppress resonant ringing. If we drive off resonance this might be mitigated, though.

Longer term plan could include multiple instantiations with microphones feeding a multichannel digitizer running at 192 kHz, and doing profiling along multiple paths.
This could evolve to acoustic tomography of dome temperature structure.

Another longer term idea would be to use the weather tower as a platform for either a microphone or speaker, and also possibly fly a UAV to measure ground layer structure.

May 2 2024.

Analog devices has a nice evaluation board that generates high voltage output with good bandwidth:

EVAL-ADHV4702-1CPZ-UG-1444.pdf

spec sheet for the op amp:

adhv4702-1.pdf

it's supposed to be able to drive 20 mA. If we run at gain of 10 there should be plenty of bandwidth available. Slew rate supports what we're trying to do.
Effective output impedance for this device is 100V/20mA ~ 5 kOhms. If impedance we're trying to drive is 1.5 kOhms then we need a turns ratio that is the square root of the impedance ratio.
That's sqrt(5 kOhms/1.5k Ohms)= 1.8 ~ 2. Primary turns are 2X secondary turns. That should maximize power transfer into the piezo.

All prices are in USD

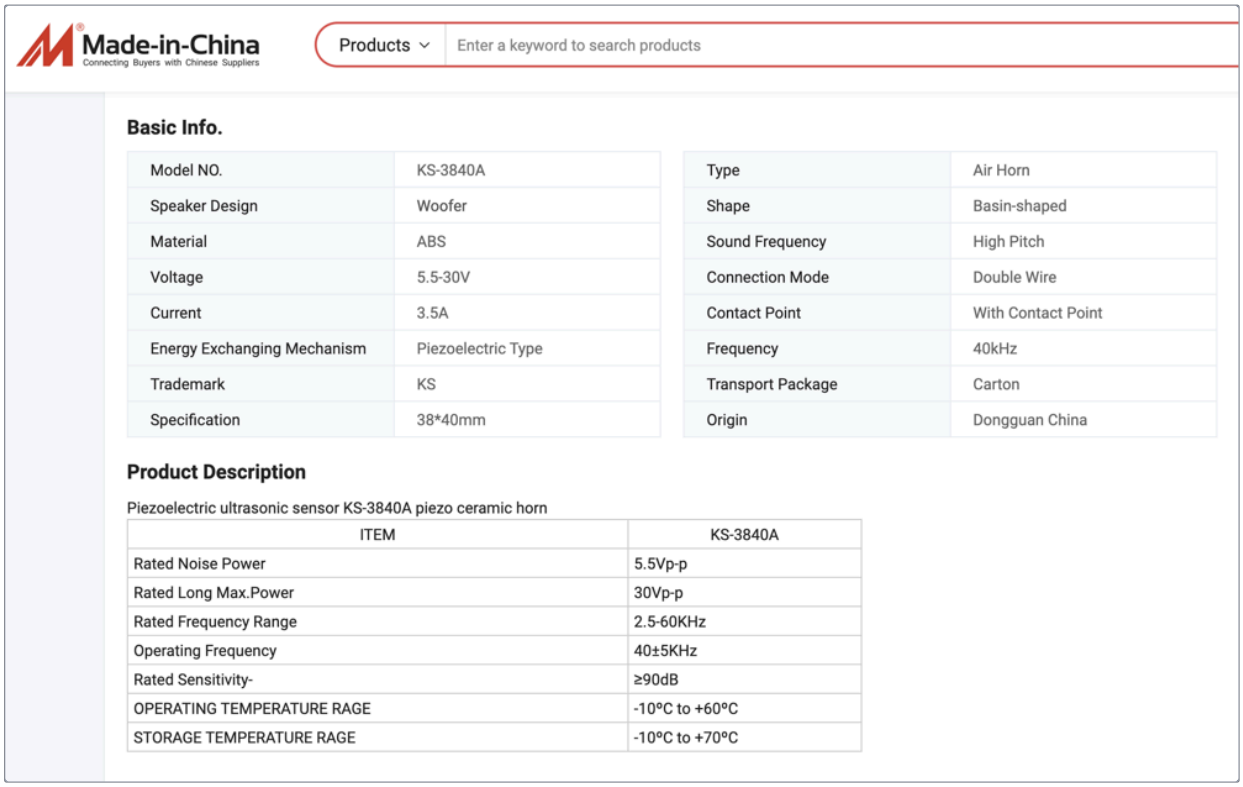
#	PRODUCT DETAILS	QUANTITY	AVAILABILITY	UNIT PRICE	EXTENDED PRICE
1	505-EVAL-ADHV4702-1CPZ-ND EVAL-ADHV4702-1CPZ EVAL BOARD FOR ADHV4702-1	2	Immediate	210.66000	$421.32

This board supplies +-100 V using only +-15V power supplies.

[toroid transformer]

May 4 2024

Small piezo speaker KS-3840A from Amazon is 78 nF capacitance, using LCR meter.



Made-in-China
Connecting Buyers with Chinese Suppliers

Products | Enter a keyword to search products

Basic Info.

Model NO.	KS-3840A	Type	Air Horn
Speaker Design	Woofer	Shape	Basin-shaped
Material	ABS	Sound Frequency	High Pitch
Voltage	5.5-30V	Connection Mode	Double Wire
Current	3.5A	Contact Point	With Contact Point
Energy Exchanging Mechanism	Piezoelectric Type	Frequency	40kHz
Trademark	KS	Transport Package	Carton
Specification	38*40mm	Origin	Dongguan China

Product Description

Piezoelectric ultrasonic sensor KS-3840A piezo ceramic horn

ITEM	KS-3840A
Rated Noise Power	5.5Vp-p
Rated Long Max.Power	30Vp-p
Rated Frequency Range	2.5-60KHz
Operating Frequency	40±5KHz
Rated Sensitivity-	≥90dB
OPERATING TEMPERATURE RAGE	-10°C to +60°C
STORAGE TEMPERATURE RAGE	-10°C to +70°C

Try looking at it with transformer. Using FT 240-31 ferrite toroid.
12 turn primary has 385 uH inductance.
~36 turn secondary (open circuit on primary) has 2950 uH inductance. Sqrt(inductance ratio) is 2.76, so close to 3.

What does piezo look like through the transformer? Is it 3X higher?
connecting speaker to secondary and connecting primary to LCR meter I see 1750 pF of capacitance, and around 150uH of inductance.

So capacitance is 22X higher. It ought to be square of turns ratio ~9 so there must be significant stray capacitance. What's the impedance at 100kHz?
|Z| is 1/omega*C = 1/(1*pi*1e5*1750E-12) = 1.8 KOhms. A 10 V drive will draw (10/1.8) ~ 5 mA. Voltage across speaker would be 30V. That's what it wants.

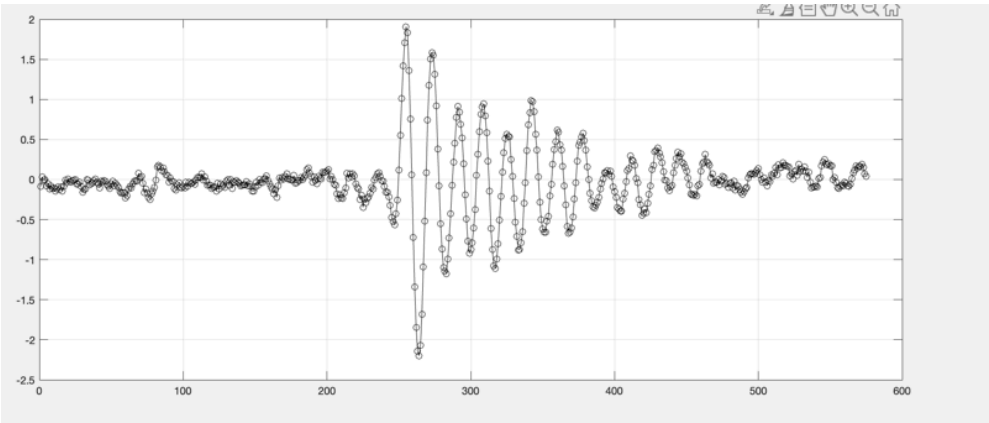
May 5, 2024

Tried out some stuff- running small piezo speaker (KS-384A) right off of waveform generator. Dodotronic 384K microphone using Audacity for data storage.

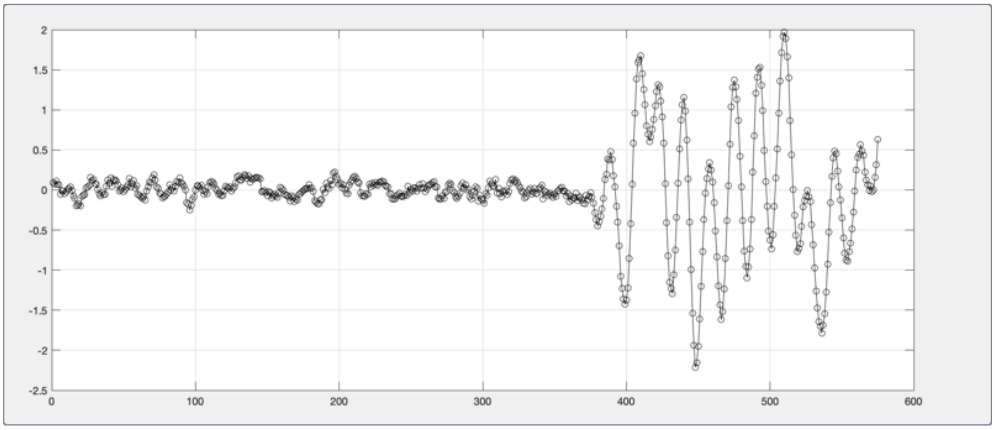
MAJOR LESSON- for audio export from Audacity use Apple file format (aiff) and 32 bit signed format. Otherwise it clips the audio levels at +-1.

set up corner reflector about 4m away. That makes a round trip travel time of about 8m/330m/s ~ 24.2 msec. To avoid pulse confusion we need to run at a pulse rep rate of under 40 Hz.
Driving with a square wave works well. It delivers positive and negative pulses so effective pulse rate is twice the function generator frequency.

Typical pulse seen by microphone upon emission. The ringing is at about 20 kHz and is presumably the natural frequency of the piezo when whacked with a square wave.

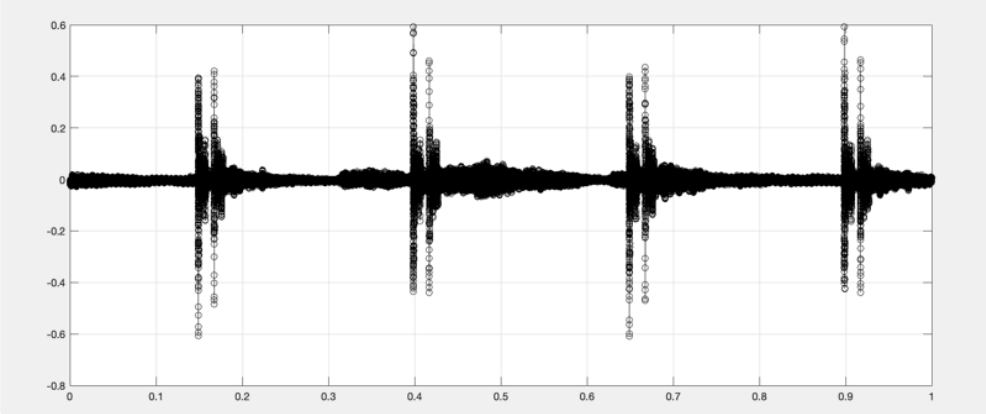


Typical reflected pulse:



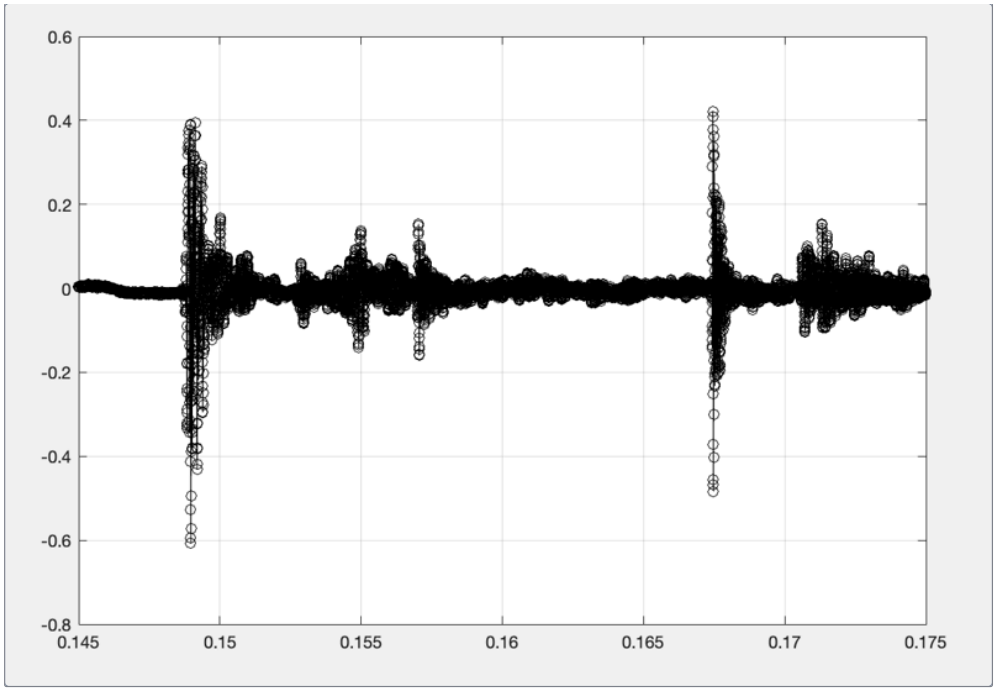
This is with 2V p-p into the piezo. Reflector was not very well isolated, lots of multipath contamination.

Let's try cranking up the amplitude to look at phase of return pulse. Also dropped square wave frequency to 2 Hz



Observations:

comparable amplitudes for start and reflected pulses. positive-going and negative-going edges are a bit different.
Here's single-pulse drive and reflection:

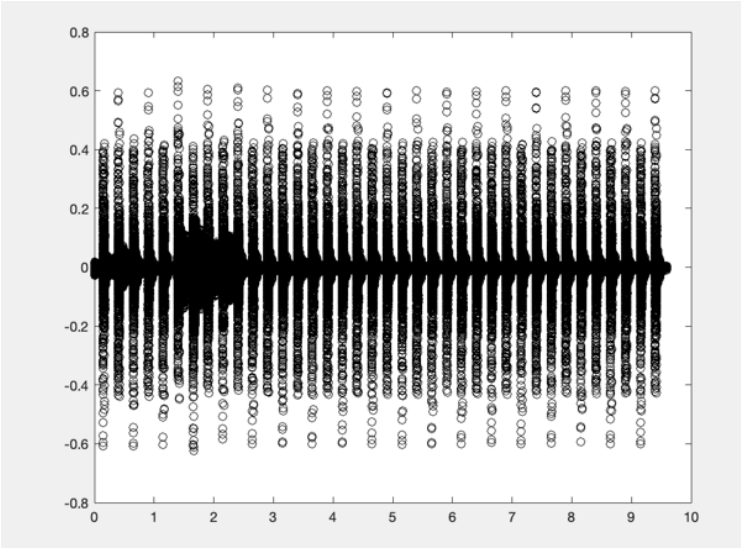


Other piezo transducers arrived today- tried to use the 175 kHz version. Shows us as around 360pF capacitance. But driving directly from function generator or through transformer coil gives zero signal. Hrrumph.
I don't understand that. Even trying to drive it on resonance at. 175 kHz give no detectable signal on microphone. Defective?

Note that driving a low frequency square wave through transformer should produce bipolar delta functions on secondary side.

Proof of principle:
Ran Amazon speaker at 2 Hz with a square wave drive 10V p-p amplitude.
recorded data with Dodtronics microphone at 384 kHz.

raw data shows some room noise that will throw things off:



wrote a really crude MATLAB program to measure round trip time delay:

```
% delays.m
% C. stubbs May 5, 2024
```

```
% assumes we have arrays t and data.

% pick an initial pulse, Nth pulse to overlay, and pulse rep rate
freq=2.00; % Hz
t1=0.1485
dt=0.0025; % plot extent in seconds
N=1;
delay=0.018; % drive-to-return delay, seconds

figure(100)
subplot(2,1,1)
plot(t,data,'ko-')
xlim([t1 t1+dt])
grid on
subplot(2,1,2)
plot(t,data,'ro-')
xlim([t1+N/freq t1+N/freq+dt])
grid on
title('multiple pulses')
shg

absdata=abs(data);

figure(200)

subplot(2,1,1)
plot(t,absdata,'ko-')
xlim([t1 t1+dt])
grid on
subplot(2,1,2)
plot(t,absdata,'ro-')
xlim([t1+N/freq t1+N/freq+dt])
grid on
title('abs value of signal')
shg

figure(300)

subplot(2,1,1)
plot(t,absdata,'ko-')
xlim([t1 t1+dt])
grid on
subplot(2,1,2)
plot(t,absdata,'ro-')
xlim([t1+delay t1+delay+dt])
grid on
title('drive and return signals')
shg

%% carve up the time series into pulses
% compute the number of data points in a half period
Npoints=floor(fs/(2*freq))

% compute number of half periods in the data set
Npulses=max(t)/(2*freq)
```

```
% carve out a succession of subsections and measure delays
maxlagtime=0.04; % max delay in seconds
maxlagindex=floor(maxlagtime*fs)

thispulse=[];
delaytimes=[];
for Npulse=1:Npoints
 startindex=(Npulse-1)*Npoints+1;
 endindex=min(startindex+Npoints+10,length(t));
 thisone=[startindex endindex]
 thispulse=absdata(startindex:endindex);

 % do autocorrelation
 [corr,lags]=xcorr(thispulse,thispulse,maxlagindex);
 lagtime=lags/fs;

 % plot it
 figure(350)
 % this smoothing scale is the period of the intrinsic piezo osc.
 plot(lagtime,smooth(corr,10),'ko-')
 grid on
 shg
 xlim([0.018 0.020])
 pause(1)

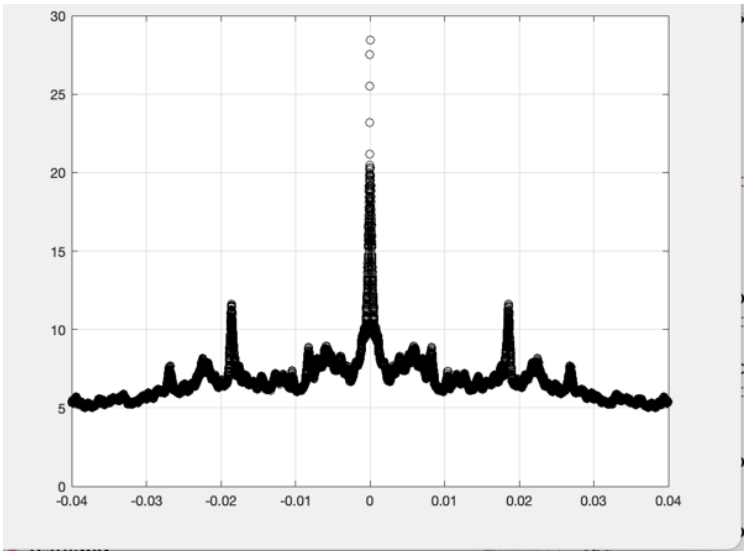
 % find corr peak, crudely, at nominal range of interest
 smoothcorr=smooth(corr,10);

 [peaks, peaklocs]=findpeaks(smoothcorr,lagtime,'MinPeakDistance',0.02);
 delaytime(Npulse)=max(peaklocs)
 pause(1)

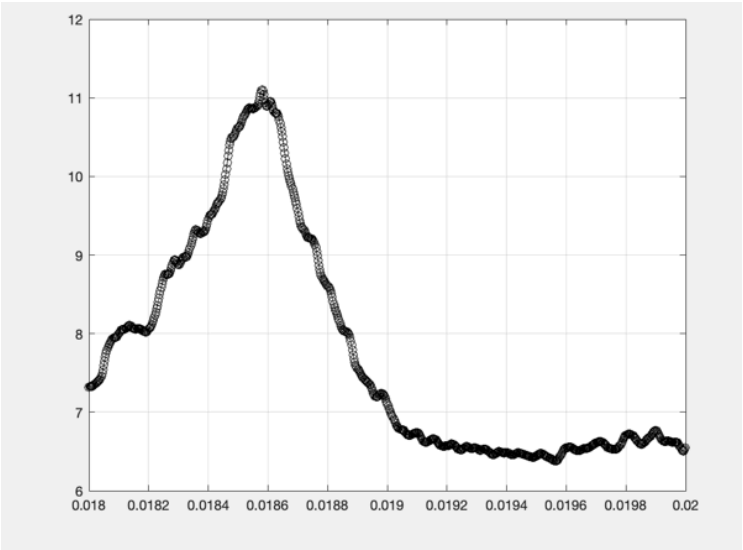
end

figure(1000)
plot(delaytimes,'ko-')
grid on
shg
```

Typical autocorrelation peak on rectified signal (accounts for phase flip on reflection):
x axis is autocorrelation lag in seconds. Peak at 20 msec is the reflection from the corner cube. Good SNR.

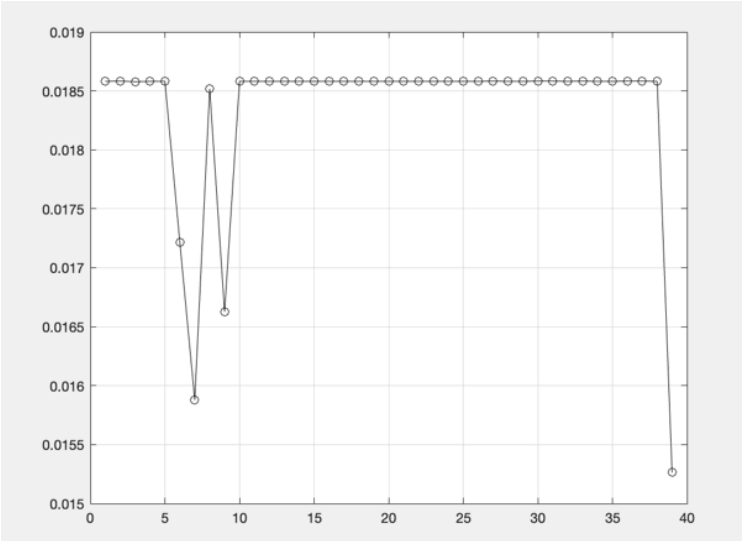


Zoom into reflection peak, after smoothing the data:



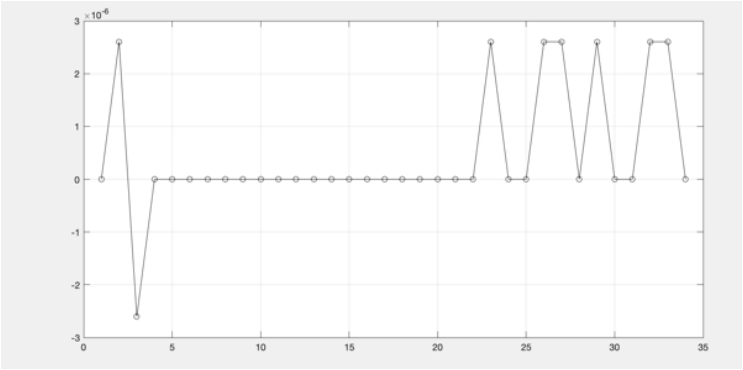
stability of return signal time delay:

round trip travel time vs. pulse number:



the outliers are either edge effects or else when there was noise in the room.

After tossing outliers, round trip seems stable to within one time bin:



Major observation- system performance is WAY better if function generator is set to 50 Ohm output impedance. This essentially puts a resistor in series with the output signal.

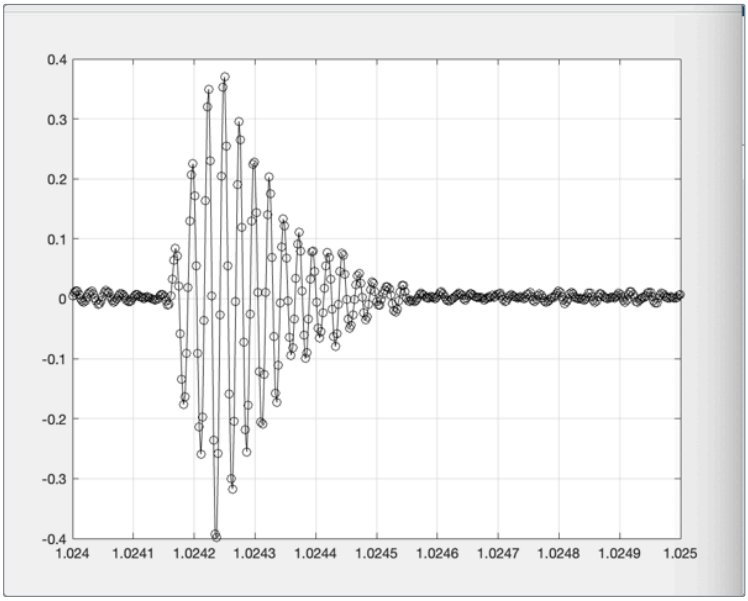
May 8, 2024

Try for a more directional emitter with large 40 kHz piezo array. Capacitance is around 90 nF, so at 40 kHz that is an impedance of |Z| = 1/omega*C = 1/(2*pi*40e3*90e-9) = 44 Ohms so reasonably matched to 50 Ohm output (except it's all reactive). Driving from function generator at 10 Hz and 50 Ohm output impedance, nothing is audible BUT get a whopping signal and nice clean return peak from reflector.

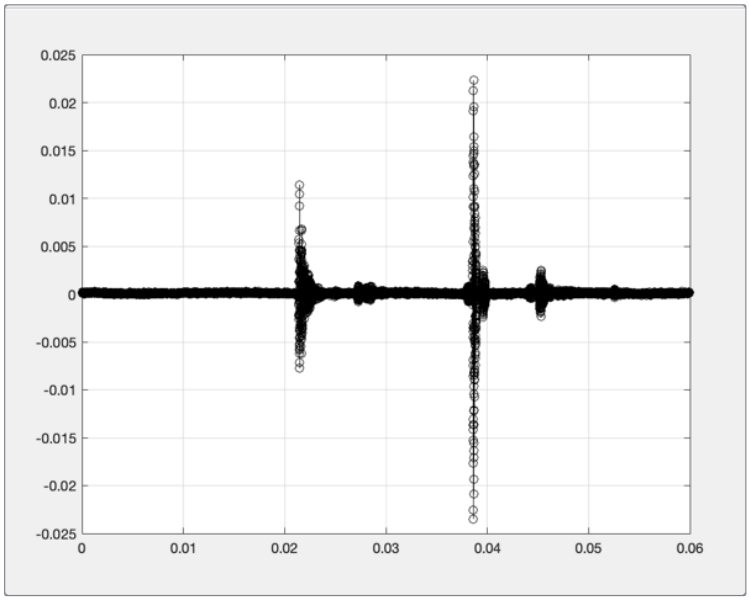


Array is 9 cm in diameter, emitting mainly at 40 kHz. Wavelength is 300m/40e3 = 8.2mm so diffraction width is lambda/9 cm ~ 8.2mm/9cm ~ 5 degrees. At 4 m away the footprint is ~ 0.4 m.

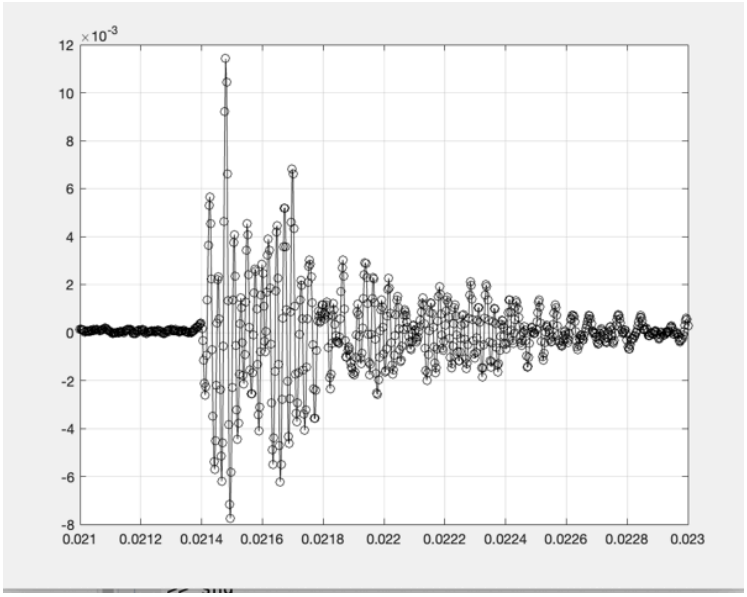
Here's the excitation signal:



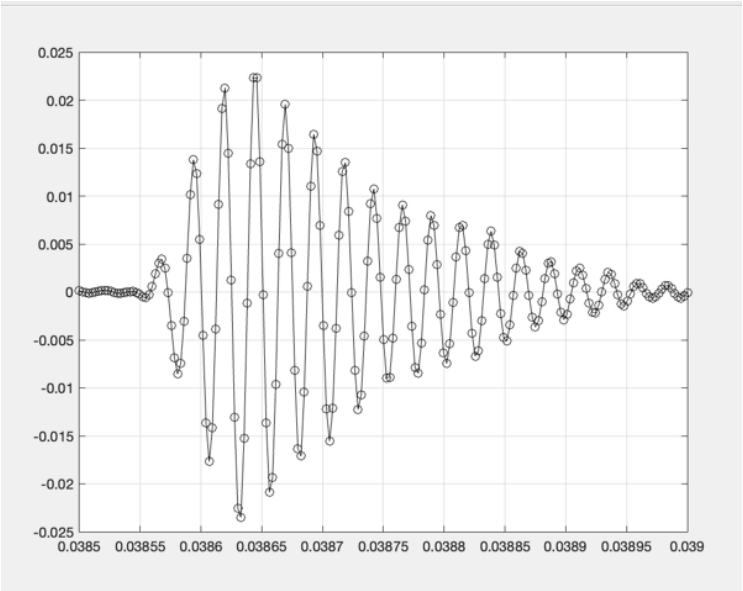
Natural freq of crystal is about 33 kHz. Now let's look for return signal. Reconfigured things a bit, here is drive (smaller) and return signal (larger)
round trip delay is 17 msec.



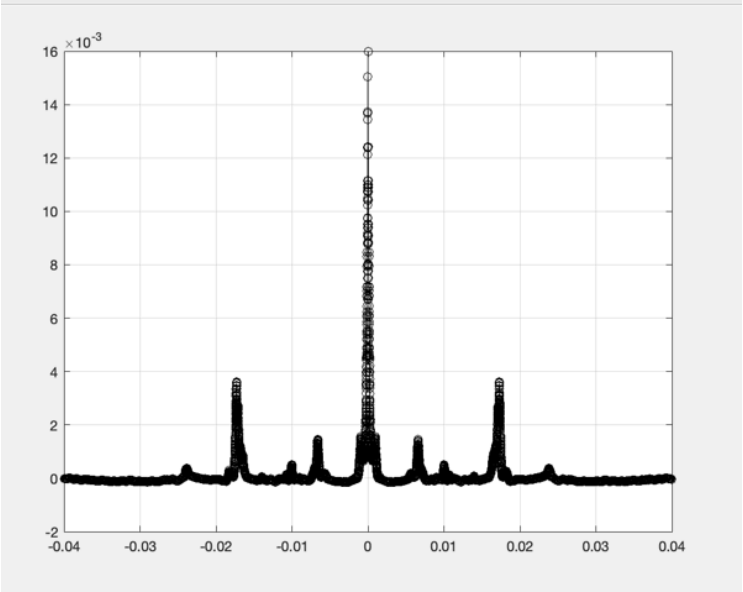
detail of drive pulse:



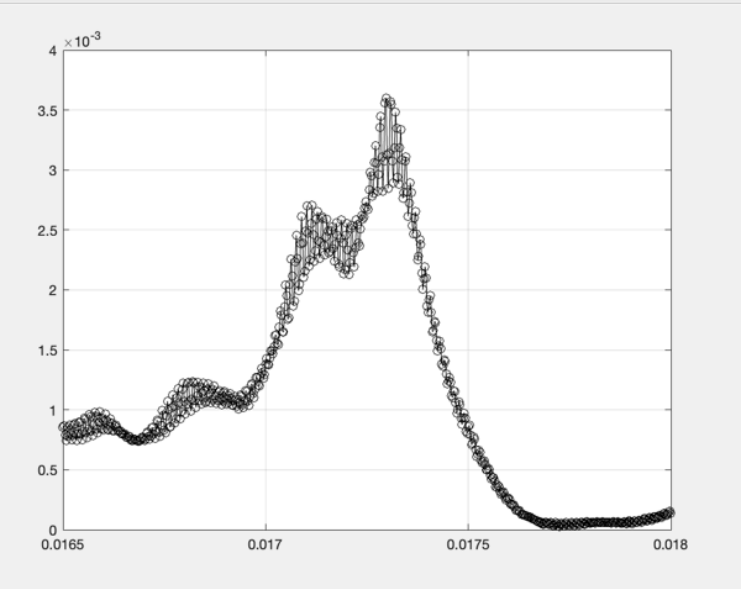
detail of return pulse:



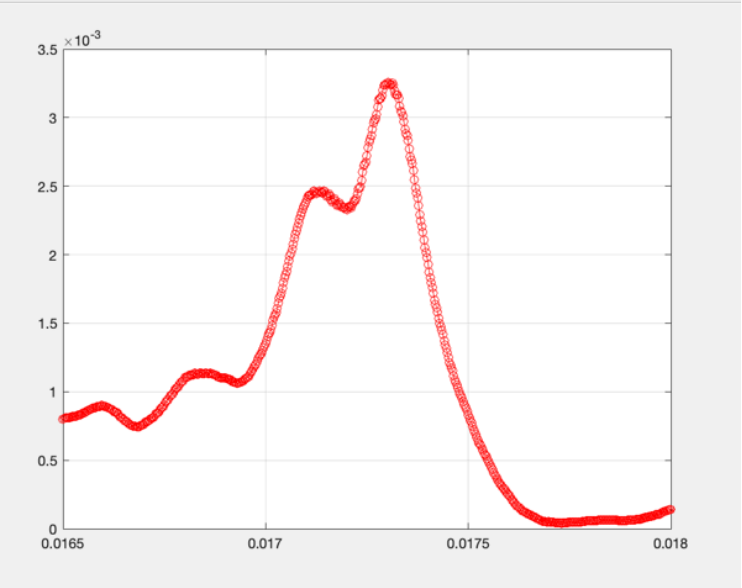
Autocorrelation plot:



Zoom into peak around reflection signal:

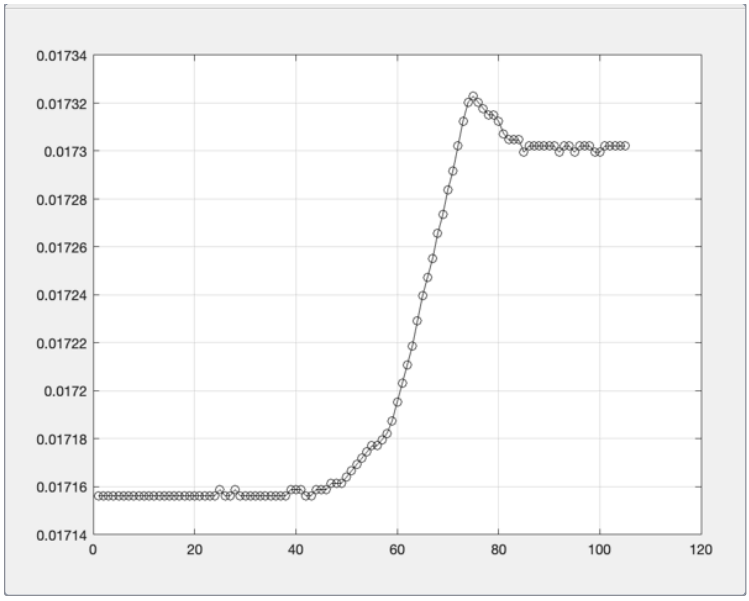


the Etaloning (oscillation) is due to beating of 33 kHz component between xmit and receive signals. We can suppress this by smoothing over one period, which is about 6 data points.
That yields this:



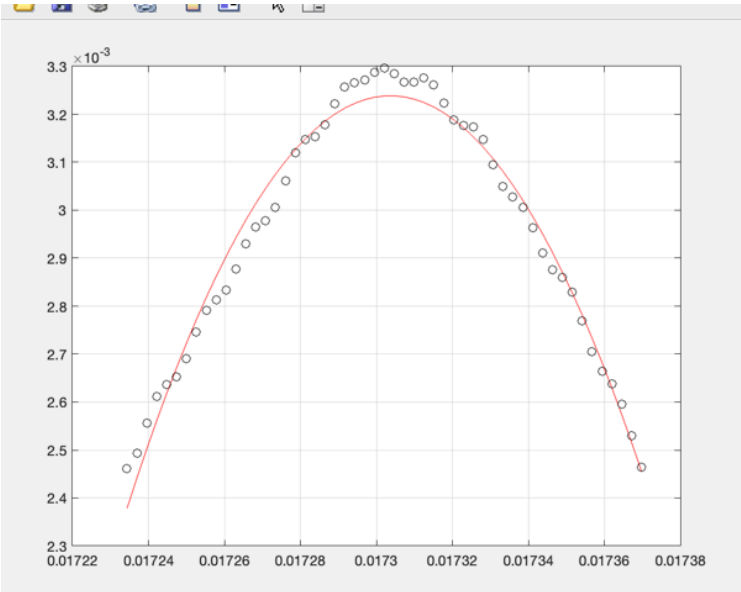
Then it's just a question of find the lag time with peak in autocorrelation function. It seems stable and repeatable to one time bin. I shifted the separation distance by hand during data collection to ensure round trip travel time also changes.

Yup:



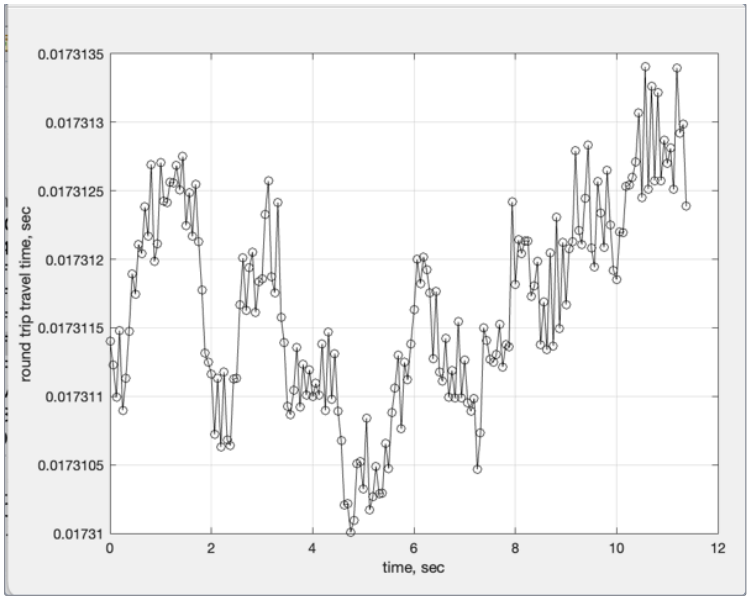
Intrinsic scatter is below 2.6 microseconds for round trip time of 17 msec, so fractional resolution is 0.1 part per thousand. **Our target (see above) was 2E-4 so this performance exceeds that goal.**

Tried a more sophisticated fit to parabola at peak, this gives finer time resolution:



Using this method I see repeatability of 0.6 microseconds for 17 msec travel time, or ~ 4e-5 fractional resolution. Exceeds measurement goal by factor of five.

Try taking a longer time series at fixed separation. Also am going to increase square wave frequency, to 8 Hz. That's 16 Hz pulse rate.



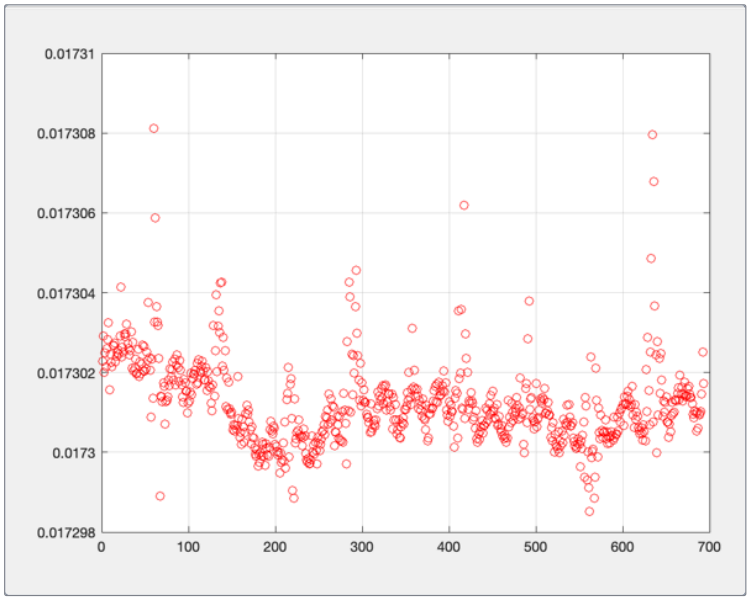
mean(peak fit value) = 0.017311666766482 at 22:51 local time
standard deviation is 7.5e-7 sec.

fractional 1 sigma scatter in travel time is (std dev)/mean = 4.3E-5.

fractional change in sound speed is 0.176% per degree. That means we are seeing equivalent temperature change dT so that sigma_T * 1.76E-5 = 4.3 E-5 so sigma_T=4.3E-5/1.76E-4 = 0.24 C

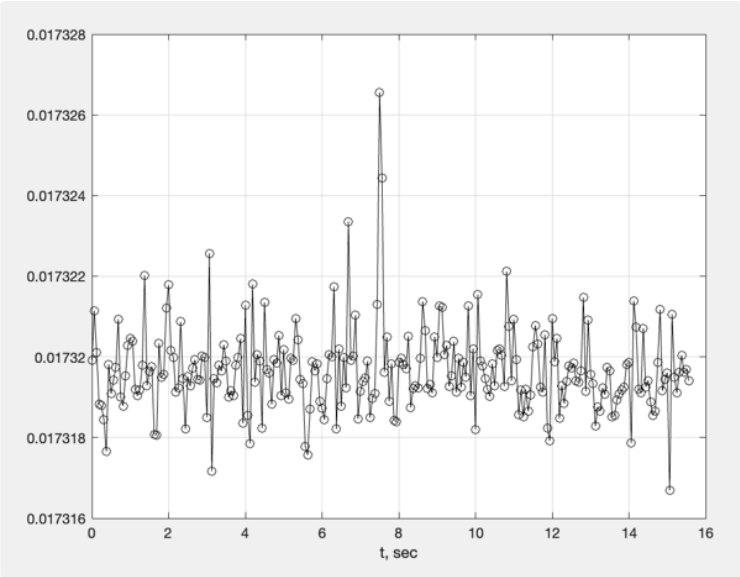
more generally, **sigma_T = 0.24 C *(std(travel time)/mean(travel time) * (1/(4.3E-5))**

mean(peak fit value) = 0.017301219231906 at 23:25 local

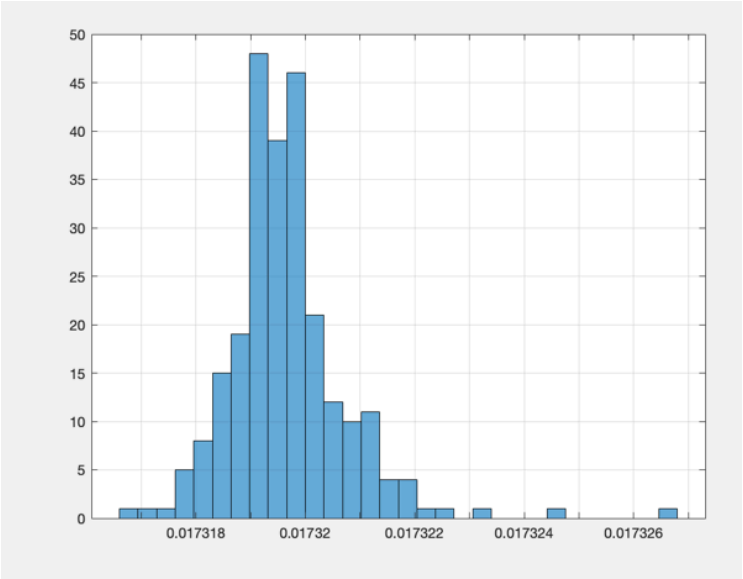


May 9.

Did another run at 5:50 am May 9
Over 16 seconds get std dev=1.07E-6
mean is 0.017319668 sec. So mean has changed by 1.8E-5, a twenty sigma change. temperature sigma is 0.35 C.



Histogram of values:

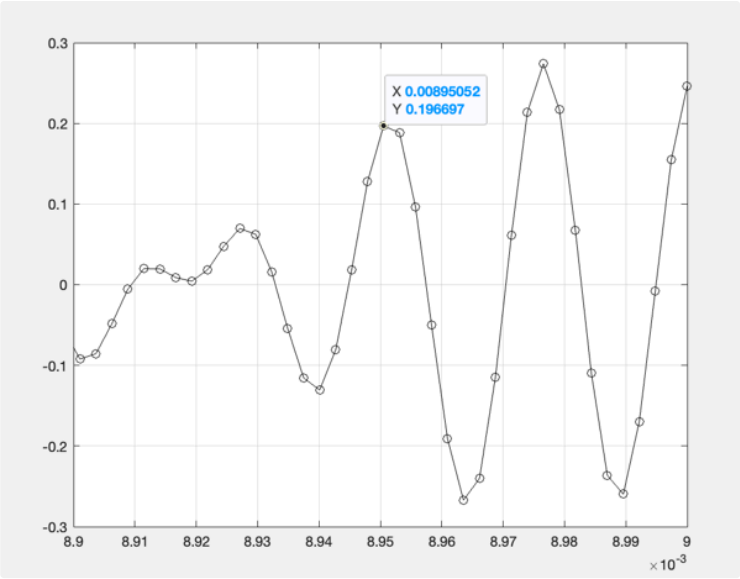


Next tests to do:

1. really short range repeatability test. Reduce drive amplitude and bring reflector much closer. Increase frequency, and look at scatter in data.
    a. did this, std deviation for 1 ms round trip travel time is 6E-7.
2. drive emitter array using the transformer coil.
3. set microphone at same plane as emitter, to get true duplication of round trip path length to suppress wind effects
4. test susceptibility to wind and to thermal perturbations

What would it take to damp oscillations in one period? here's drive pulse detail, freq of piezo is 35 kHz. Period is 28 microseconds.

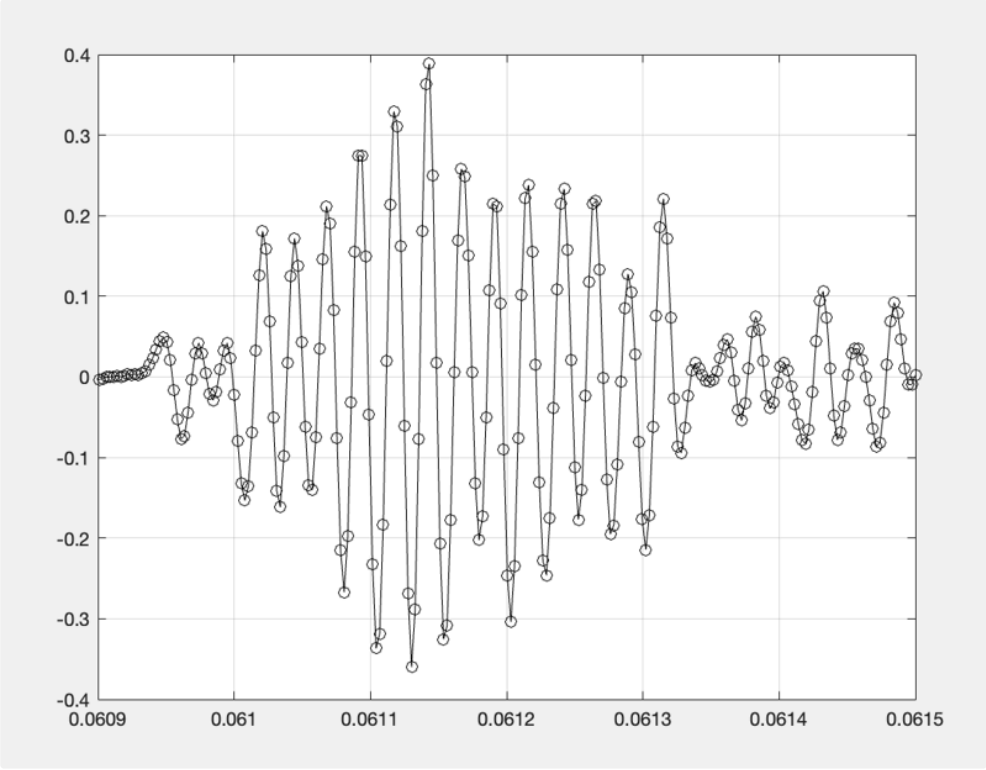
This plot is no coil and no parallel resistor, 50Ohm drive impedance.



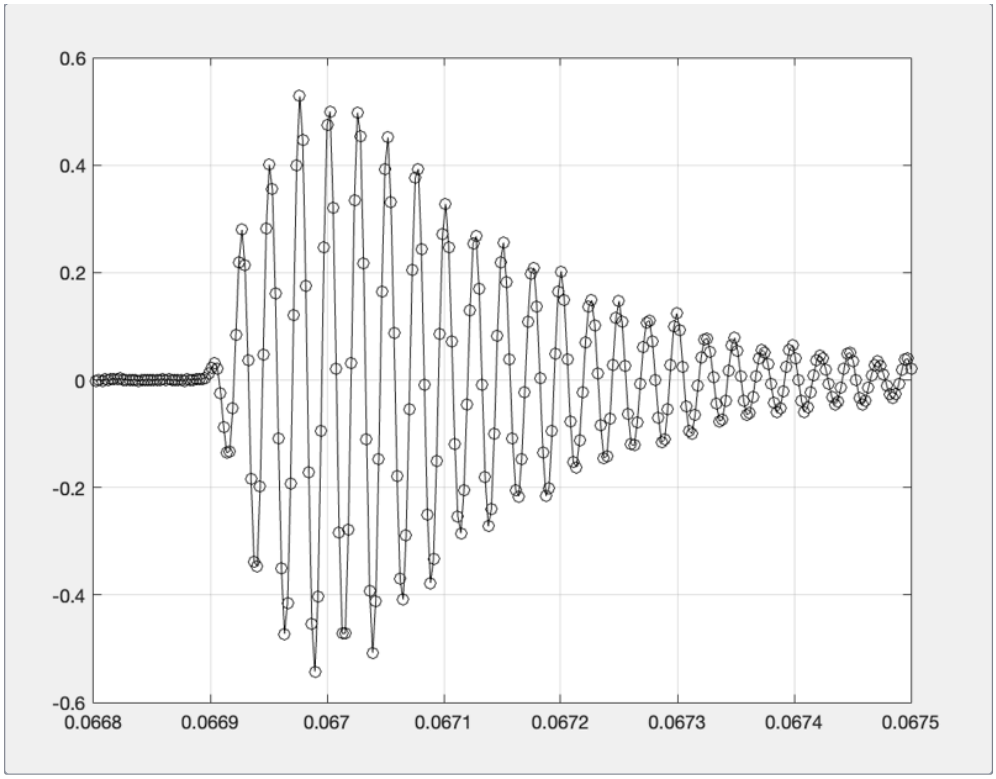
50 Ohms in parallel seems to completely short it out? Now it doesn't seem to work at all? Hard to understand how this could have damaged it... no. Just need to restart Audacity for data colleciton.

If we put a resistor in parallel with the piezo array, will that dissipate the oscillations? Power transfer is maximized if impedances are matched so try around 50 Ohms.

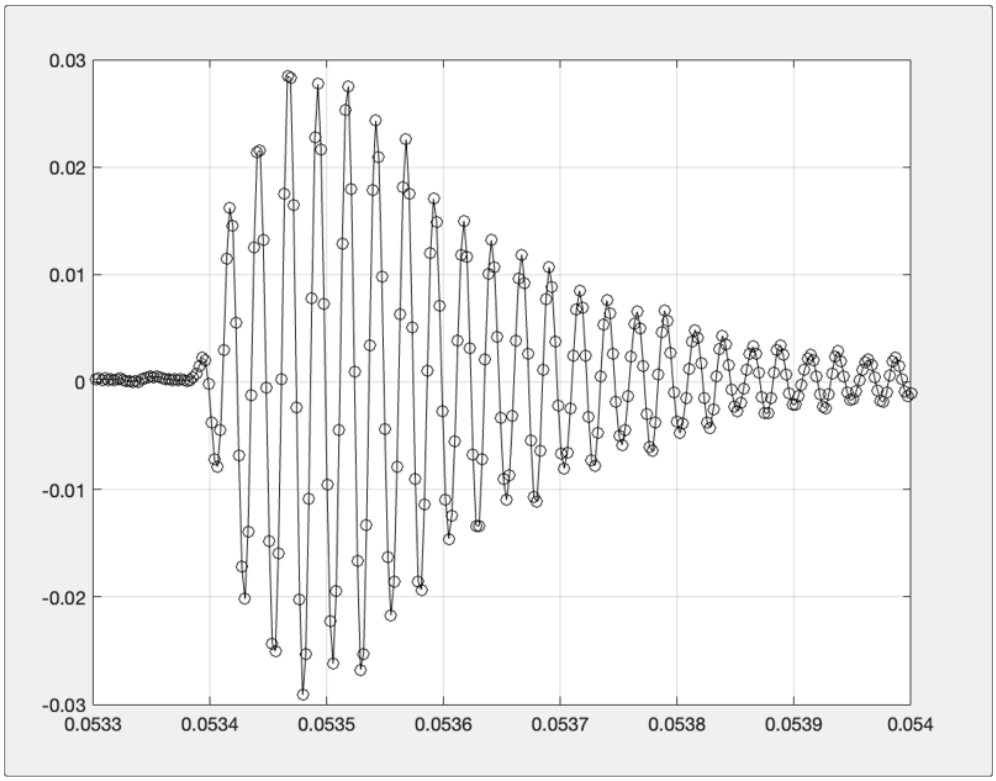
Here is no coil, no damping resistor, get around 13 cycles at 35 kHz.



Here is through coil, no resistor:



I think that looks cleaner for correlation analysis. How about adding damping resistor? Using 50 Ohms:



Doesn't reduce ringing much. So conclusion is that coil-drive is better. Seems to run fine from 0.1V p-p amplitude too.

Repeat long range testing but now using coil drive.

standard deviation values are bimodal. Upgoing and downgoing square wave edges seem to differ. So that can circumvented by putting in half the drive freq in the analysis code. That makes it look at only one edge polarity. Std dev for 38 msec travel time is a whopping 6E-5. That's 10 sigma compared to short-range repeatability test values.

making lean system involves using a small function generator.

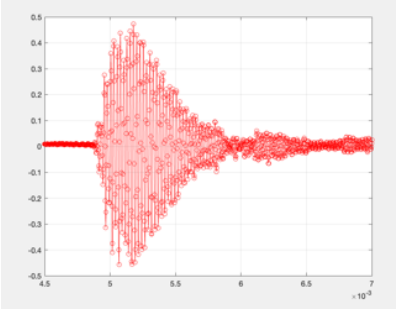
Data sheet is here: UTG900E Datasheet.pdf

Period of transient oscillations is 28.64 microseconds. If we let it go through one cycle then have opposite sign of drive to suppress oscillations, we want pulse width to be 1.5 periods or 85.94 microseconds. Let's try that. Not much change. What if we explore different number of half-cycle pulse widths?

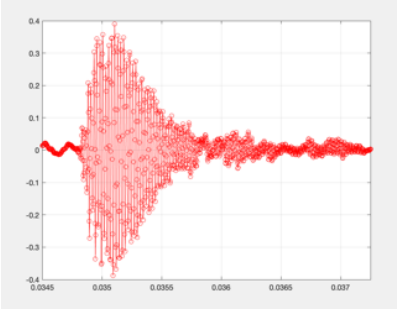
pulse duration in microseconds vs half-periods:

28.6458
57.2917
85.9375
114.5833
143.2292
171.8750
200.5208

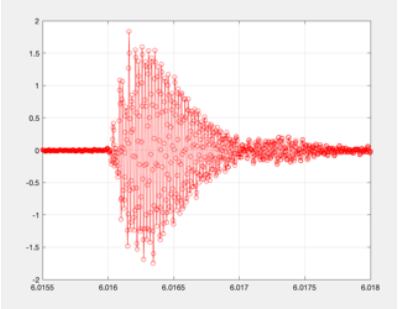
14.0 usec pulse duration.

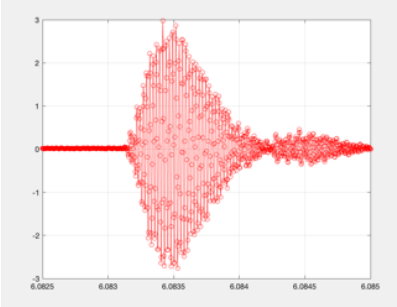


28.64 pulse duration

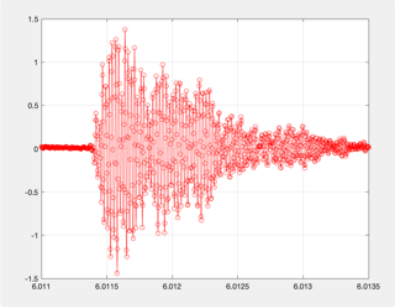


114.58 microsec pulse duration:





5 microsec pulse



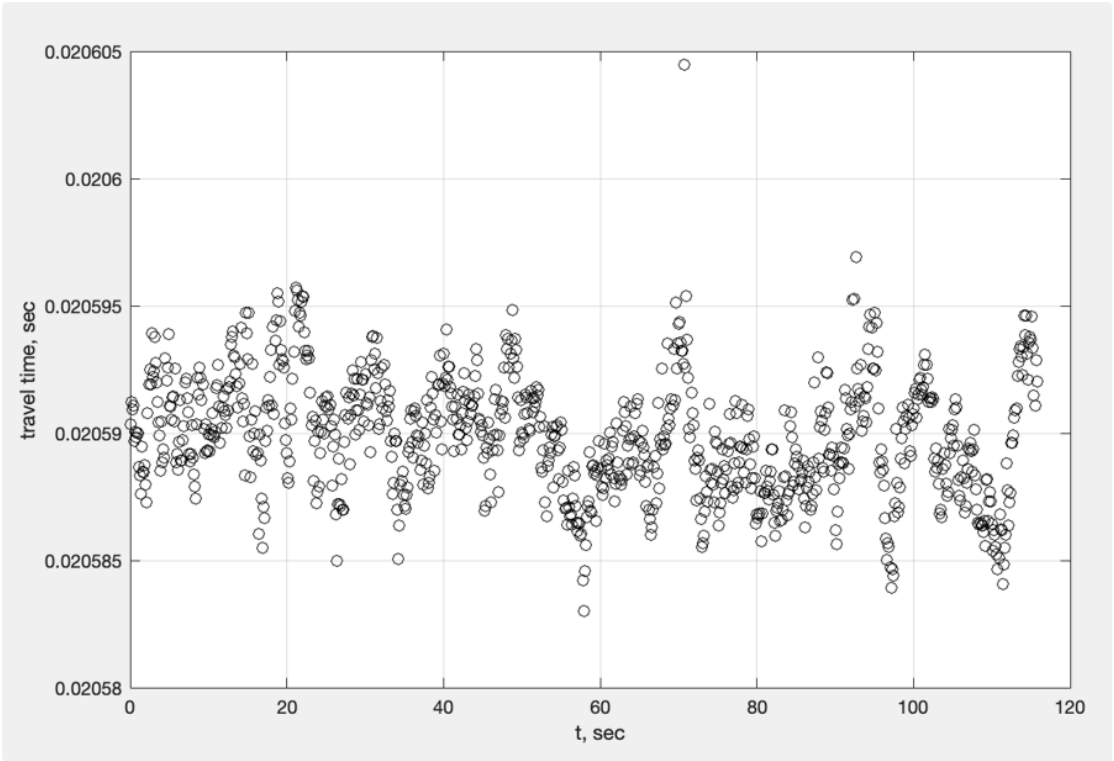
There isn't a big effect here, but 14 usec looks best to me. Longer pulses have more amplitude, but we can get that back with gain. Shorter pulse train is better.

Made a revised version of the analysis program, that imposes high pass audio filter at 10 kHz and works well with pulse excitation rather than using square waves.

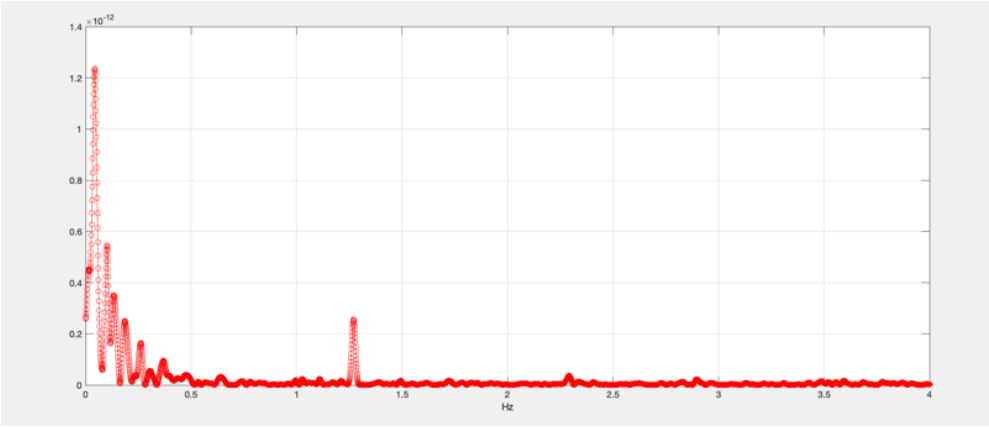
function generator settings:



Two minutes of data taken in living room with 6.8m round trip path length.



What is the power spectrum of these variations? Huh, weird:



OK. This basically works. UT932E function generator, driving piezo array though toroid transformer. No parallel load resistor. Dodotronics 384kHz microphone. Radar reflector. Laser pointer for alignment.
delayV2.m analysis code: delaysV2.m

Packing up!

May 14, 2024, at Cerro Pachon

We set up BATPING in Rubin control room. Round trip travel time to reflector is 59 msec, so total path length is 330m/s * 50e-3 s = 19.5m (!)
Set up pointing with laser pointer, running 40 kHz emitter array off function generator with 50 Ohm output impedance, through transformer coil. Return signal SNR is really high.
Kane Sjoberg suggested a test of putting hot kettle of water under the beam path. Wasn't sure what to expect... what we saw was total annihilation of the return off the corner cube, although reflection from wall on the far side was still there. So this perturbation clearly put us into the strong-scattering regime, where scintillation is v. large. I modified delay2.m to include storing strength of autocorrelation signal. If xmit pulse is constant, this is a measure of return echo strength.

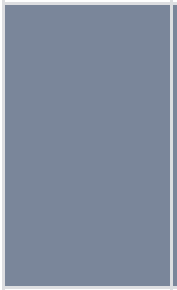
Drive configuration: tries to wait 1 period before negative going falling edge, to suppress transients, but that doesn't work too well...
Getting good stability in control room tests.



Set up reflector on tripod in AuxTel dome. Delay time is 22 msec so path length is 7.3 meters.

There is a lot of acoustic noise from dome rattling but using high pass filter of 10 kHz gets rid of most of it. We get around 2.8e-4 msec standard deviation of travel time, so way more than we see in benign control room.

time (local Chilean time)	dome status	data file	duration	time delay std dev	kept fraction	correlation strength std dev	notes
2040	closed	DomeTest2.aiff	35 sec	2.8E-4 sec	0.86		
2104	closed	DomeTest3.aiff	62 sec	2.3E-4 sec	0.84		
2110	closed	DomeTest4.aiff	105 sec	1.2E-4	0.82		possible peak ambiguity in autocorrelation?
2123	closed	DomeTest5.aiff	58 sec	2.8E-4	0.76		are we losing pulses because of pointing drift?
2140	closed	DomeTest6.aiff	60 sec	3.0E-4	0.71		losing an increasing fraction of pulses, I think because the trigger start pulse is getting lost in noise from dome? Try measuring separation between echo peaks, set time delay to 1/4 Hz = 250 msec and set period to 2 Hz so we get 2 return echos per subsample. This is of course susceptible to drift in the function generator but I expect that to be slow variation?

							Woah. That seems to work really well. Redo the analysis setting pulse rep rate to 2 Hz and prior on delay to 250 msec. The standard deviation we get from this should be sqrt(2) larger than what we'd see for a single round trip delay scatter. Interestingly, this way of doing things is independent of distance to reflector as well.
							Analysis changed here to scatter of interval between echos. WAY less noisy.
2040	Closed	DomeTest2	35 sec	1.16E-6	0.91	0.953	
2104	Closed	DomeTest3	62 sec	1.90E-6	0.99	1.61	
2110	Closed	DomeTest4	105 sec	1.81E-6	1.0	1.13	
2123	Closed	DomeTest5	58 sec	2.20E-6	1.0	1.00	
2140	Closed	DomeTest6	60 sec	1.80E-6	0.95	1.02	bistability in this one? seems to be toggling between adjacent data point bins?

2213	Closed	DomeTest7	60 sec	2.42E-6	1.0	1.27	Note that the standard deviation is one sample time interval on A/D converter....
2214	Closed	DomeTest8	60 sec	2.35E-6	1.0	1.15	
2220	Closed	DomeTest9	60 sec	1.15E-6	1.0	1.36	sonic anemometer std dev is 0.07, which is a typical low value. Sonic temperature is 9C.
2235	Closed	DomeTest10	60 sec	2.21E-6	1.0	1.30	
2237	Closed	DomeTest11	60 sec	2.25E-6	1.0	0.94	now cracking dome open 2%. weather is bad and wind speeds are high so only partially opening.
2255	cracked open 2%	DomeTest12	60	1.835E-6	1.0	1.07	sonic anemometer std dev is bouncing around bit no real change we can see. Dome vents are off.
2301	cracked open 2%	DomeTest13	60	2.015E-6	1.0	1.30	considering opening dome more, but note wind speeds are high so being conservative.
2316	cracked open 2%	DomeTest14	5=60	2.33E-6	1.0	1.48	
2323	cracked open 2%	DomeTest15	60	2.30E-6	1.0	1.56	

	cracked open 2%						reflector was bumped. oops.
	cracked open 2%	DomeTest16	60	1.81E-6	0.992	9.5 (2.5 when fixed)	one weird data point threw off scint.
	cracked open 9%	DomeTest17	60	1.58E-6	0.9917	1.9	at 2327 dome opened to 9%
2340	cracked open 9%	DomeTest18	97 sec	1.6E-6	0.9897	1.82	
2345	cracked open 9%	DomeTest19	300 sec	1.88E-6	0.99	2.01	

May 15, 2024

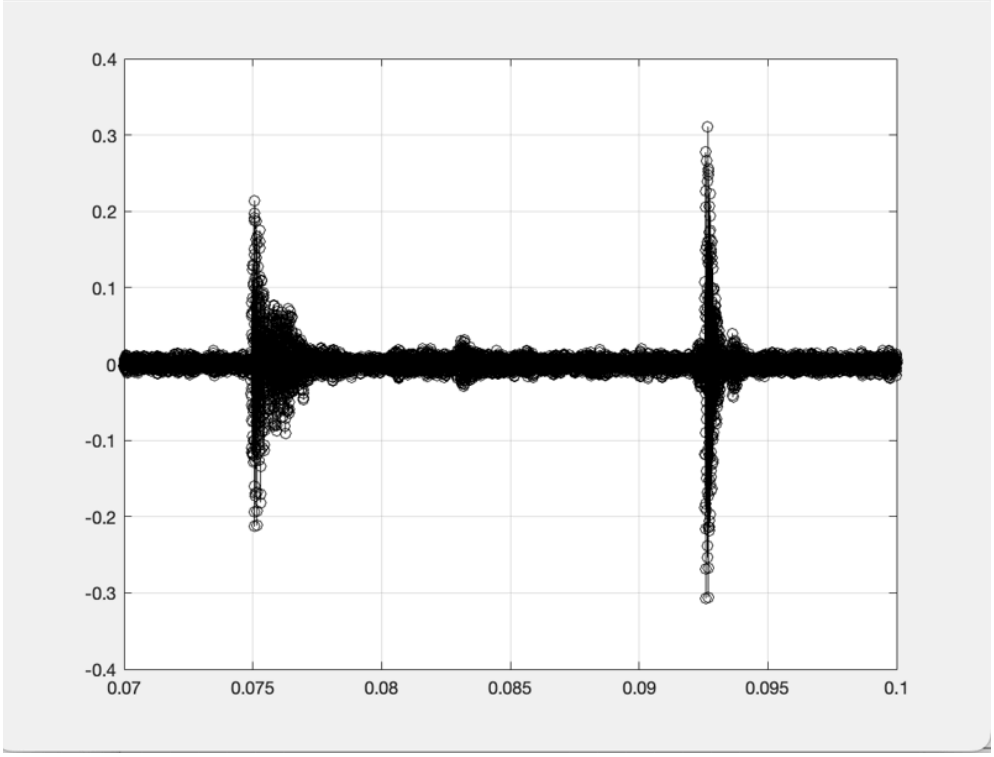
did a test run at 0925, still aligned.

echo-to-echo scatter is 1.7E-6.
Partial dome opening last night did make a slight change in anemometer standard deviation, went up to perhaps 0.14C of standard deviation.
We've convinced ourselves that the scatter in the echo-to-echo time delay is indicative of changes in line integral of temperature



May 15 2024

1345 local
removed the conical baffle from the front of the BATPING microphone. We now get a much better trigger pulse from the array:



DomeTest100.aif

looking at echo-echo delay:

std = 1.43E-6
keptfraction=0.77 ❓scintscatter=0.42

DomeTest100-DomeTest104 are all dome closed.
Now try w dome open. T=5.48C on anemometer.
1520 local opening slit

DomeTest105 is during dome opening

1525 local dome is now open, taking DomeTest106
1528 anemometer standard deviation is now 1.2 deg!!
temp has fallen to 3 deg C

DomeTest107 through 111 are with slit open
Humidity is high so closing now at 1535.

DomeTest114 onwards is with dome slit closed again. Temperature is around 2.5C
DomeTest116 is 4 min long.

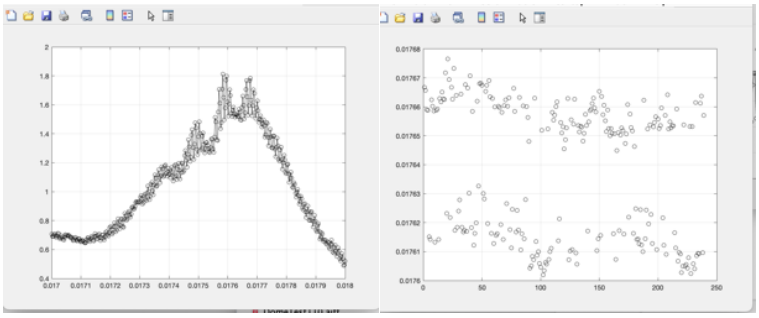


Made two analysis programs: one does echo-echo time delay, the other does trigger-echo delay.

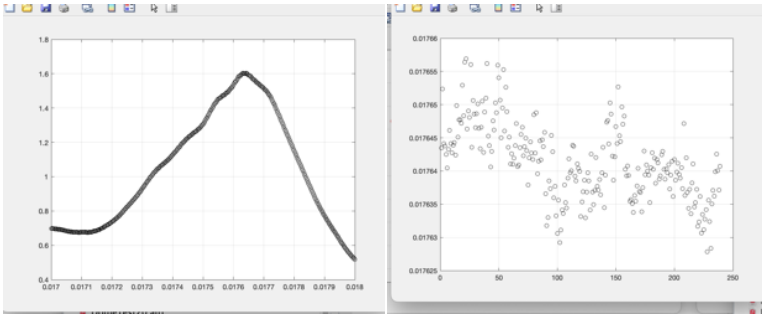
file	time (local)	dome	anemometer temp	anemometer std	EchoEcho delay sigma (microsec)	EchoEcho corr sigma	TrigEcho mean delay (ms)	TrigEcho mean delay long smoothing	TrigEcho delay sigma (microsec)	TrigEcho delay sigma w long smoothing	TrigEcho corr sigma	keptfract TrigEcho	notes
DomeTest100	1336	closed	6.06	0.20	1.51	0.43	17.562	17.5783	14.5	8.1	0.10	0.98	
DomeTest101	1457	closed	5.59	0.20	1.62	0.56	17.569	17.5182	5.2	7.3	-	0.81	
DomeTest102	1458	closed	5.45	0.20	1.73	0.49	17.575	17.5980	4.0	5.5	0.19	0.92	
DomeTest103	1500	closed	5.45	0.20	1.83	0.62	17.588	17.6040	5.9	7.5	0.34	0.75	
DomeTest104	1501	closed	5.53	0.20	1.73	0.52	17.593	17.6042	4.2	6.4	0.19	0.65	
DomeTest105	1524	opening	3.19	0.68	1.93	0.60	17.623	17.6267	7.0	18 but rapid change over interval	0.53	0.84	

DomeTest106	1526	open	2.89	1.22	1.76	0.56	17.667	17.6796	26.1	14.8	0.23	1.0	bistability in trig-echo data
DomeTest107	1527	open	2.30	1.17	1.76	0.56	17.667	17.6697	26.1	9.0	0.24	1.0	same file saved twice? Yes.
DomeTest108	1530	open	1.75	1.16	1.76	0.50	17.659	17.6872	25 but 8.1 after cleaning up bistability	15.2	0.19	1.0	
DomeTest109	1531	open	1.75	1.05	1.76	4.7	17.674	17.6872	contaminated	15.4	0.26	0.99	
DomeTest110	1532	open	1.68	1.02	1.84	0.74	17.663	17.6669	contam	18.4	0.27	0.79	
DomeTest111	1534	open	2.50	0.65	1.56	0.55	17.681	17.6869	contam	15.6	0.23	1.0	
DomeTest112	1535	open	3.17	0.62	1.64	0.37	17.688	17.6975	contam	11.3			
DomeTest113	1539	closing	3.35	0.47	1.64	0.37		17.6975					another dup file? Yes
DomeTest114	1541	closed	4.31	0.47	1.67	0.50	17.649	17.6535	contam	7.1	0.27	1.0	
DomeTest115	1543	closed	4.74	0.27	1.50	0.38		17.6415		5.7			
DomeTest116	1548	closed	4.88	0.22				17.6263					
DomeTest117													

This is the bistability that is screwing up the trig-echo measurements. First plot is autocorrelation vs. lag time, second is peak found.

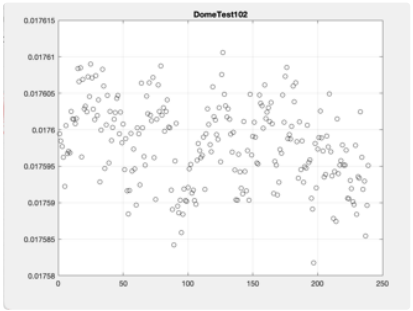


Increased the smoothing scale over the autocorrelation function before finding the peak, improved things:

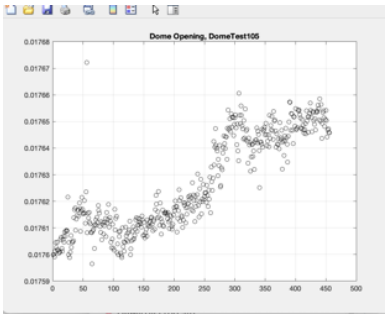


re-running Trig-Echo delay code with this change, added relevant columns above.

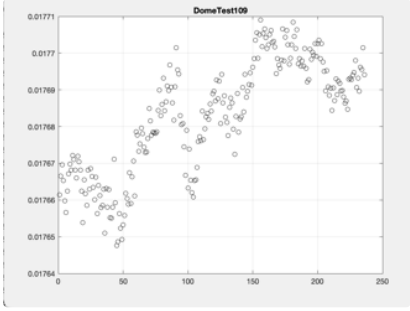
DomeTest102 is dome closed:



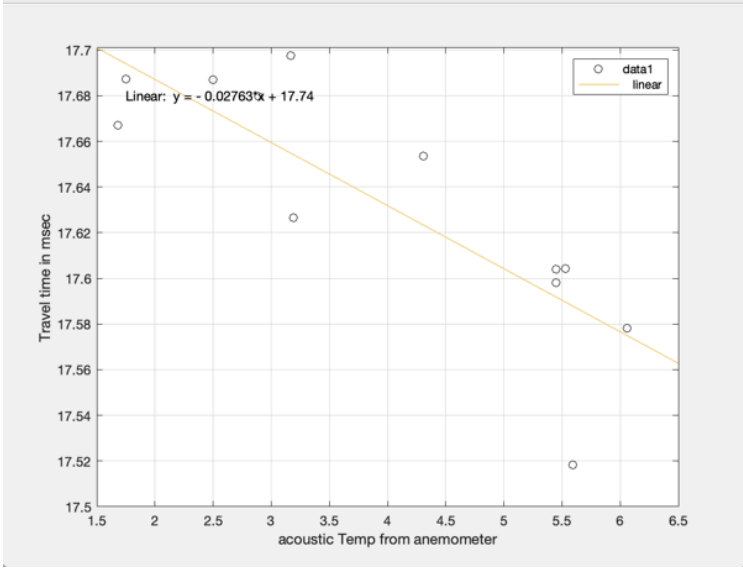
DomeTest105 is during dome opening and air cooling:



DomeTest109 is with dome open and air cooling:

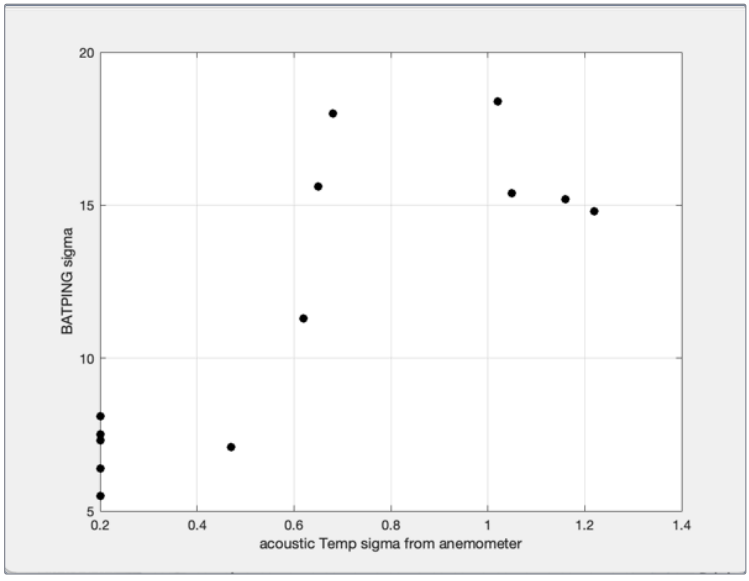


Plotting travel time vs. temperature:



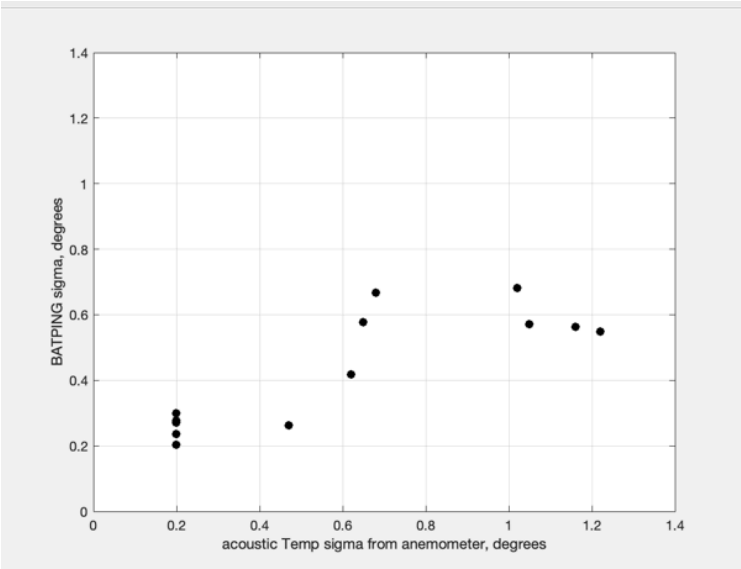
Fit gives -0.027 msec per degree. That's a fraction change of - 1.5 E -3 compared to theory value of -0.6/330 = -1.8 E -3. Fractional error in that determination is 16%.

What about variances?



Convert scatter in time delay to scatter in temperature: Our sigmas are in microseconds. Slope of line above is 27 microseconds per degree. So Batping sigmaT = Batping sigma(time)/27 where sigma(time) is in microseconds.

replot with temp vs. temp:



Status and summary, May 15, 2024.

Overall, echo determination has good sensitivity and performance. There are artifacts coming from the 40kHz self-resonance of the sensor. But there are two main puzzles:
1) why isn't our signal in the last plot above huge? Am convinced the standard deviation should scale as sqrt(number of uncorrelated volumes). The commercial anemometer sees as much as one degree RMS over a really small volume, ours should be ~10 times higher, but it isn't, we aren't getting nearly the 'gain' I was expecting. In fact the sigmas are a factor of two lower?

2) why isn't there a bigger signal in the echo-to-echo channel? It seems rock-stable at the pulse rep rate. Could we have latched onto the trigger pulse? I don't think so.

One interpretation of what we're seeing- the integral of the temperature just doesn't vary much between adjacent pings, even though the anemometer sees lots of power at 10 Hz. We should look at travel time variations over longer time scales. One way to do that is to look at the delay jitter between much more widely separated pulses.

Next-gen system goals:
We are strongly limited by the piezo response being dominated by 40 kHz resonance, that propagates nicely in air. Using one of the other emitters and driving it off resonance should avoid the ringdown problem, but at the expense of less efficient coupling of energy into the air. A paper from Analog Devices describing driving piezos off resonance is here: driving-audio-piezoelectric-transducers.pdf
IF we drive a delta function into a higher-freq-self-resonance device, the air will act as a low pass filter and we should get back a broadened pulse but with minimal ringdown. We will likely have to drive it pretty hard though.

List of things to take to Chile for in-dome testing:

laptop
Dodotronics microphone
function generator
coax cable
toroid(?)
reflector
clamps and mounting stuff

Some data files:
DomeTest102.aiff
DomeTest109.aiff

References

726.pdf		
atmosphere-10-00460-v3.pdf		
an5522-ultrasound-behavior-and-guidelines-of-analog-imp23absu-mems-microphone-stmicroelectronics.pdf		
Dokhanchi_Dissertation_ATOM.pdf		
Ultramic384_EVO_User_Guide.pdf		
07B1 ACOUSTIC TRAVEL TIME TOMOGRAPHY FOR SIMULTANEOUS INDOOR AIR TEMPERATURE AND AIRFLOW MEASUREMENTS.pdf		
Recent Progress in Acoustic Travel-Time Tomography of the Atmosph.pdf		
stae871.pdf		
 LeCroy waveform generator user manual,		
 LeCroy Programming manual		